

Miloslav Pavelka, Ing.

Lead Software Security Architect · Secure software design and delivery

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15 years of experience and enthusiasm in building secure software. Deeply invested in secure software design and delivery, well-versed in security architecture and threat modeling, and experienced as a software engineer, team leader, security auditor and penetration tester.

EXPERIENCE

Lead Software Security Architect

Kooperativa (Vienna Insurance Group) · May 2026 – Present

- **AI-assisted threat modeling:** Introduced, as my own initiative, a process that turns threat-modeling session transcripts into a threat register for every delivery (5+ new services released to production each month), built on a threat catalog I maintain. The hard part is judgement: picking the threats that matter, and keeping track of those that can't be addressed yet instead of dropping them.
- **Security engineering team:** Building the team from scratch, including hiring and establishing its processes, to deliver tooling that addresses security design issues in bulk; currently leading 4 engineers.
- **Security tooling at scale:** Designed software and CI/CD tooling for security controls, monitoring and quality gates, covering all 600+ active GitLab projects; run the organization-wide Secure Software Development Life Cycle (SSDLC) program.
- **Identity & access management:** Designed and oversee delivery of integrated IAM solutions covering authentication, RBAC, ABAC and ReBAC.

Software Security Architect

Kooperativa (Vienna Insurance Group) · 2024 – May 2026

Financial institution processing high volumes of sensitive organizational and customer data.

- **Application security architecture:** Designed and approved secure solutions at the architecture and code level, for in-house and vendor-provided software, with threat modeling and secure code review; identified security gaps in the existing ecosystem and designed patches and target solutions. Stack: Java, Kotlin, JavaScript (React, Angular), Go, Python, WSO2, Kafka, Camunda, Redis, MSSQL, Kubernetes, Helm, Terraform, Ansible.
- **Secure SDLC:** Authored and implemented the internal SSDLC policy and co-authored standards; rolled out Aqua Security for SAST, Kubernetes workload protection and cloud security posture management (CSPM).
- **Enterprise security architecture:** PAM, IAM/IDM, Active Directory, Entra ID; network security and segmentation (Palo Alto, F5); endpoint security and vulnerability scanning (CrowdStrike, Tenable).

Senior Security Engineer

Hotjar (Contentsquare) · 2021 – 2024

- **Authorization platform:** Designed and built the policy decision point (PDP) that all ~50 microservices depended on for authorization. Stability, reliability and scalability were the core design principles, with caching to avoid unnecessary round trips.

- **Security software engineering:** Designed and built the user verification and authentication components used across the products, meeting strict security requirements for data in use, in transit and at rest. Implemented SAML and OAuth 2.0. Stack: Python 3, TypeScript, Node.js, Kafka, Elasticsearch, Docker, Kubernetes, AWS, GitHub Actions; applied cryptography (AES, RSA, ECC, HMAC, PBKDF).
- **Cloud security:** Built IAM/PAM for new and existing cloud assets with Terraform (IaC) and Jenkins; integrated third-party SSO providers (Auth0, Okta).
- **Security assessment & triage:** Triaged external penetration test findings and bug bounty reports and supported the owning teams with fixes; ran secure architecture and code reviews and red team activities.
- **Security culture:** Led the 5-person security chapter and published internal security awareness resources.

Senior Security Engineer, Consultant, Team Leader

TeskaLabs Ltd · 2015 – 2021

- **Security software engineering:** Co-created Certificate Authority (CA) and Public Key Infrastructure (PKI) systems for the X.509 and C-ITS (cooperative intelligent transport systems) standards, built on a microservice architecture with distributed data stores protected by KMS and HSMs. Co-created an ISO 27001-compliant log management product. Stack: C, Python (Cython for compute-heavy parts), Kotlin, Java, Nginx, MongoDB, Kafka, Elasticsearch, Docker, Azure.
- **Team leadership:** Managed a distributed team of engineers from the Czech Republic and Croatia that built a point-of-sale system for Android devices.
- **Security audits, penetration testing & training:** Alongside the product work, audited technology and processes for ~5 customers a year, with white-box and penetration testing following OWASP; ran hands-on security workshops for R&D teams.

Software Engineer

DHL IT Services · 2014 – 2015

- Developed applications used across DHL corporate entities: a PKI system for IoT certificates on courier devices, a scanner configuration tool with X.509 certificate management, and a Lync telephony number assignment portal. Linux, Python, Apache, Active Directory (LDAP).

Web Developer

Freelance · 2012 – 2014

- Learned what it takes to sell a service the hard way.

SKILLS

- **Security architecture:** threat modeling (STRIDE, PASTA, MITRE ATT&CK, OCTAVE), C4 model, OWASP, NIST SP 800-160 Vol. 2 Rev. 1, ISO 27001:2022 Annex A, risk management
- **Identity & access:** SAML, OAuth 2.0, JWT, Kerberos, NTLM, Basic Auth; RBAC, ABAC, ReBAC; PAM
- **Cryptography & PKI:** AES, 3DES, RSA, ECC, HMAC; X.509, KMS, HSM; secrets management
- **Secure engineering:** secure coding, structured audit logging, redundancy and high availability, vulnerability management
- **Languages & APIs:** Python, Java, Kotlin, TypeScript (senior); Go, Swift, C++, Ruby, PHP (familiar); REST, GraphQL, multipart
- **Platform & delivery:** AWS, Azure, Docker, Kubernetes, Terraform; CI/CD (GitHub Actions, GitLab CI, Travis); Sentry, Datadog, Elasticsearch, Kibana
- **Assessment & testing:** Burp Suite, ZAP, Nmap, Metasploit, Nessus, Snyk, Semgrep, Android and iOS emulators, Chromium DevTools
- **Regulations:** GDPR, NIS2, SOC 2, HIPAA (general understanding)
- **Leadership:** careful listening, giving and receiving feedback, leading by example while delegating responsibility

EDUCATION & CERTIFICATION

- ISC2 CISSP
- **Master’s degree (Ing.), University of Economics in Prague:** studied while already working in software development and cybersecurity; the school shaped my leadership skills.

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